hit you one after another. In sound, these are pressure waves, and higher frequency is higher pitch.

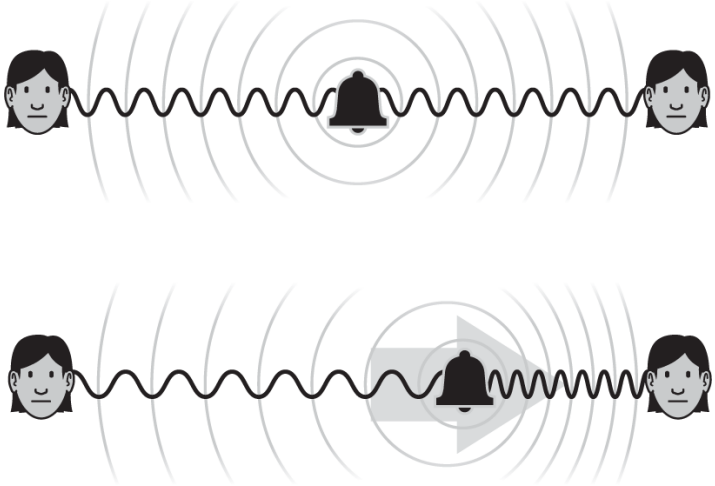


Figure 8: Illustration of a Doppler shift. When the source of the sound is stationary, the frequency heard by two stationary observers is the same. When the source of the sound is moving, the sound is stretched out to lower frequencies for the observer the source is receding from and compressed to higher frequencies for the observer the sound is approaching. The former hears a low note, while the latter hears a high one.

It turns out that light does something similar. A light moving toward you quickly will shift to a higher frequency, and one moving away will shift to a lower frequency. For light, frequency corresponds to color, so this shift looks like a color change. The electromagnetic spectrum extends far beyond the visible, but as a shorthand, when a Doppler shift of light occurs, a shift up is called a *blueshift* (because high-frequency visible light is on the blue end of the spectrum) and a shift down is called a *redshift*. Extremely blueshifted visible light could get all the way to gamma rays, while extremely redshifted light would show up as a radio signal. This phenomenon is one of the most important and versatile tools in astronomy, as it allows us to see, just from the color of a star or a galaxy, if it's moving toward us or away.

Of course, in practice, it's not quite that simple. (Astrophysics can be frustrating that way.) Some stars and galaxies are just redder, inherently, than others. How, then, to know if something is red because it's just red, or red because it's receding?[fn2] The key is that the light is never just one